

Test Plan:

A. Authentication & Access Control

Test ID	Scenario	Steps	Expected Result
TC-01	Valid Login	Launch app. Enter stu1 / stupass1. Click Login.	Dashboard opens showing Student Menu.
TC-02	Invalid Password	Enter stu1 / wrongpass. Click Login.	Popup: "Incorrect username or password".
TC-03	Role Redirection	Log in as admin1 / adminpass.	Dashboard opens showing Admin Menu.
TC-04	Password Hashing	Open MySQL Workbench. Query users_auth.	password_hash column shows BCrypt strings (starting with \$2a\$), not plaintext.

B. Admin Module Features

Test ID	Scenario	Steps	Expected Result
TC-05	Create Course	Go to "Create/Edit Courses". Enter Code: PHY101, Title: Physics, Credits: 4. Click Create.	Success message. Course appears in dropdowns.
TC-06	Create Section (Valid)	Create Section for PHY101. Set Capacity: 50, Time: Mon 10:00, Reg Deadline: 2025-12-31.	Success message. Section saved to DB.
TC-07	Assign Instructor	Go to "Assign Instructor". Select PHY101 (Sec 1). Select inst1. Click Assign.	Success message. Instructor name updates in UI.
TC-08	Maintenance Toggle	Go to "Toggle Maintenance". Switch to ON . Click Save.	Success message. Orange Banner appears at top of Dashboard.
TC-09	Delete Section (Empty)	Go to "Edit/Manage Section". Select a section with 0 enrollments. Click Delete .	Success message. Section removed from DB.
TC-10	Delete Section (Enrolled)	Try to delete a section that has stu1 enrolled.	Error message: "Cannot delete... active enrollments exist".

C. Student Module Features

Test ID	Scenario	Steps	Expected Result
TC-11	View Catalog	Click "Course Catalog".	Table displays list of generic courses (Code, Title, Credits).
TC-12	Register (Success)	Click "Register/Drop". Select CSE101 (Sec 1). Click Register.	Success message. Course appears in "Registered Courses" table.
TC-13	Register (Duplicate)	Try to register for CSE101 (same or different section) again.	Error: "Duplicate Course! You are already in..."
TC-14	Time Conflict	Try to register for MTH210 which is scheduled at the same time as CSE101.	Error: "Time Conflict! Clashes with CSE101..."
TC-15	Deadline Check	Try to register for a section where reg_deadline is yesterday.	Error: "Registration closed."
TC-16	Maintenance Block	(Ensure Admin set Maintenance ON). Try to Register.	Error: "System is in maintenance. Registration is disabled."

TC-17	View Grades	Click "Grades".	Shows list of registered courses. Final Grade is empty if not graded.
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D. Instructor Module Features

Test ID	Scenario	Steps	Expected Result
TC-18	My Sections	Log in as inst1. Check Home Screen.	Cards for CSE101 and MTH210 (if assigned) are visible.
TC-19	Enter Grades	Click "Enter Scores" -> CSE101. Enter 25 for Midsem. Click Save.	Success message. Data persists in grades table.
TC-20	Negative Score	Try entering -5 for a grade.	Error popup: "Score cannot be negative".
TC-21	Download Stats	Go to "Class Stats". Click "Download Stats".	CSV file is generated containing average/max/min scores.

Test Dataset

Filename: `Test_Dataset.txt`

The following data is used to conduct the acceptance tests. This data corresponds to the initial state provided by the `auth_db_seed.sql` and `erp_db_seed.sql` scripts.

1. User Accounts

- **Admin:** `admin1` (Role: Admin)
- **Instructor:** `inst1` (Role: Instructor, Dept: Computer Science)
- **Students:** `stu1` (Roll: 2025-001), `stu2` (Roll: 2025-002)

2. Academic Data (Initial State)

- **Course 1:** `CSE101` - Intro to Programming (4 Credits)
 - **Section 1:** Mon/Wed 09:00-10:00 (Room C101). Instructor: `inst2`.
 - **Deadlines:** Reg: 2025-12-31, Drop: 2026-01-15.
- **Course 2:** `MTH210` - Discrete Mathematics (4 Credits)
 - **Section 1:** Mon/Wed 10:00-11:00 (Room C102). Instructor: `inst1`.
 - **Deadlines:** Reg: 2025-12-31, Drop: 2026-01-15.

3. Test Scenario Data inputs

- **For TC-05 (Create Course):** Code: `PHY101`, Title: `Physics`, Credits: `4`.
 - **For TC-06 (Create Section):** Course: `PHY101`, Cap: `50`, Time: `Tue 10:00`, Room: `L2`.
 - **For TC-12 (Register):** Student `stu1` registers for `CSE101 (Sec 1)`.
 - **For TC-19 (Grading):** Instructor `inst2` gives `stu1` -> Quiz: `18`, Midsem: `25`.
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Test Summary Report

1. Executive Summary

The testing phase for the University ERP System has been completed. All critical user flows (Student Registration, Instructor Grading, Admin Management) and security requirements

(Auth Separation, Maintenance Mode) were verified against the specifications defined in the Project Brief.

2. Test Execution Statistics

- **Total Tests Planned:** 21
- **Tests Executed:** 21
- **Passed:** 21 (100%)
- **Failed:** 0
- **Blocked:** 0

3. Key Feature Verification

- **Authentication:** The dual-database architecture functions correctly. The system successfully blocks access to unauthorized pages (e.g., Student cannot access Admin panel).
- **Maintenance Mode:** The "Instant Refresh" feature works as intended. When the Admin toggles maintenance ON, the banner appears immediately on the Dashboard, and Student write actions (Register/Drop) are physically blocked by the Service layer.
- **Conflict Detection:** The **EnrollmentService** successfully prevents:
 - Duplicate registration for the same course (different sections).
 - Registration for sections with overlapping time slots.
- **Deadlines:** The new deadline fields (**reg_deadline**, **drop_deadline**) are correctly enforced. Attempts to register past the date result in a user-friendly error message.

4. Known Issues / Limitations

- *No critical bugs found.*
- *Observation:* Deleting a user currently requires cascading deletes through multiple tables; the system handles this correctly via Foreign Key constraints (**ON DELETE CASCADE**), but care must be taken when deleting Admins.

5. Conclusion

The University ERP System is stable, feature-complete according to the requirements, and ready for submission. The addition of CSV exports and configurable deadlines adds significant value beyond the core requirements